

8(a)(i)	any three from: 1 detect stimuli ; 2 receptor potential ; 3 impulse / action potential, in <u>sensory</u> neurone ; 4 <i>ref. to chemoreceptors</i> on (moth) antenna ; 5 act as transducers ;	3 CON receptor cell confused with receptor protein once then ECF 2 ALLOW generator potential 3 DO NOT ALLOW pass, receptor potential / stimulus, to sensory neurone 5 ALLOW description e.g. convert energy from one form to another / convert (named) stimulus energy to electrical energy
8(a)(ii)	any three from: 1 <u>switch gene</u>, on / off ; 2 transcription factor ; 3 allows / stops, <u>RNA polymerase</u> to, act / bind ; 4 gene for <u>enzyme</u> to make, lipid / scent (chemical) ; 5 AVP ;	3 IGNORE gene expression 1 ALLOW <u>gene switching</u> 2 ALLOW activator / repressor 5 e.g. light detected / light receptor / phytochrome e.g. clock genes e.g. negative feedback e.g. mRNA / protein, degradation (→ daily cycle)
8(b)	A: mRNA ; B: cDNA / ssDNA ; C: (ss DNA) probe / reporter / oligo ;	3 A: ALLOW messenger RNA B: ALLOW complementary / single-stranded, DNA C: IGNORE complementary